1.23 e and Natural Logarithms

Worked Solutions

1 a)

$e^x = 8 \Rightarrow x = \ln 8.$

1 b)

$e^{2x} = 4 \Rightarrow 2x = \ln 4 \Rightarrow x = \frac{1}{2}\ln 4.$

1 c)

$e^{3x} = 8 \Rightarrow 3x = \ln 8 \Rightarrow x = \frac{1}{3}\ln 8.$

1 d)

$e^{5x} = \frac{1}{100} \Rightarrow 5x = \ln \frac{1}{100} \Rightarrow x = \frac{1}{5}\ln \frac{1}{100}.$

2 a)

$e^{4x+3} = 9 \Rightarrow 4x + 3 = \ln 9 \Rightarrow x = \frac{1}{4}(\ln 9 - 3).$

2 b)

$e^{5x-2} = 10 \Rightarrow 5x - 2 = \ln 10 \Rightarrow x = \frac{1}{5}(\ln 10 + 2).$

2 c)

$e^{9-2x} = 4 \Rightarrow 9 - 2x = \ln 4 \Rightarrow 2x = 9 - \ln 4$
$\Rightarrow x = \frac{9}{2} - \frac{1}{2}\ln 4.$

2 d)

$e^{1-5x} = \frac{1}{5} \Rightarrow 1 - 5x = \ln \frac{1}{5} \Rightarrow 5x = 1 - \ln \frac{1}{5}$
$\Rightarrow x = \frac{1}{5} - \frac{1}{5}\ln \frac{1}{5}$
$(= \frac{1}{5} - \frac{1}{5}\ln 5^{-1} = \frac{1}{5} + \frac{1}{5}\ln 5).$

3 a)

$\ln x = 7 \Rightarrow x = e^7.$

3 b)

$\ln x = -5 \Rightarrow x = e^{-5}.$

3 c)

$\ln x = \frac{1}{3} \Rightarrow x = e^{\frac{1}{3}}.$

3 d)

$\ln x = -\frac{2}{5} \Rightarrow x = e^{-\frac{2}{5}}.$

4 a)

$\ln 4 + \ln 5 + \ln 6 = \ln 20 + \ln 6 = \ln 120.$

4 b)

$\ln 8 + \ln 3 - \ln 4 = \ln 24 - \ln 4 = \ln 6.$

4 c)

$\ln 12 - \ln 4 - \ln 6 = \ln 3 - \ln 6 = \ln \frac{3}{6} = \ln \frac{1}{2}.$

4 d)

$\ln 40 - \ln 8 + \ln 5 = \ln 5 + \ln 5 = \ln 25.$

5 a)

$\ln x + \ln 2 = \ln 8 \Rightarrow \ln 2x = \ln 8 \Rightarrow 2x = 8 \Rightarrow x = 4.$

5 b)

$\ln x - \ln 5 = \ln 3 \Rightarrow \ln \frac{x}{5} = \ln 3 \Rightarrow \frac{x}{5} = 3 \Rightarrow x = 15.$

5 c)

$\ln 8 + \ln x = \ln \frac{1}{3} \Rightarrow \ln 8x = \ln \frac{1}{3} \Rightarrow 8x = \frac{1}{3} \Rightarrow x = \frac{1}{24}.$

5 d)

$\ln 50 - \ln x = \ln \frac{3}{2} \Rightarrow \ln \frac{50}{x} = \ln \frac{3}{2} \Rightarrow \frac{50}{x} = \frac{3}{2} \Rightarrow x = \frac{100}{3}.$

6 a)

$\ln e = 1.$

6 b)

$4 \ln \frac{1}{e} = 4 \ln e^{-1} = -4 \ln e = -4.$

6 c)

$\ln \frac{1}{e^5} = \ln e^{-5} = -5 \ln e = -5.$

6 d)

$\ln \sqrt[3]{e} = \ln e^{\frac{1}{3}} = \frac{1}{3} \ln e = \frac{1}{3}.$

7 a)

$2 \ln 2 + 2 \ln 4 = \ln 4 + \ln 16 = \ln 64.$

7 b)

$3 \ln 2 - 2 \ln 3 = \ln 8 - \ln 9 = \ln \frac{8}{9}.$

7 c)

$2\ln 9 + 3\ln 3 - 2\ln 6 = \ln 81 + \ln 27 - \ln 36 = \ln \frac{81\times 27}{36} = \ln \frac{243}{4}$.

7 d)

$2\ln 10 - 4\ln 2 - 2\ln 5 = \ln 100 - \ln 16 - \ln 25$
$= \ln \frac{100}{16\times 25} = \ln \frac{1}{4}$.

8)

$3\ln x - 4\ln y = \ln x^3 - \ln y^4 = \ln(\frac{x^3}{y^4})$.

9 a)

$4\ln p + 2\ln q + 3\ln r = \ln p^4 + \ln q^2 + \ln r^3$
$= \ln(p^4 q^2 r^3)$.

9 b)

$3\ln p + 5\ln q - 4\ln r = \ln p^3 + \ln q^5 - \ln r^4$
$= \ln(\frac{p^3 q^5}{r^4})$.

9 c)

$6\ln p - 2\ln q - 8\ln r = \ln p^6 - \ln q^2 - \ln r^8$
$= \ln(\frac{p^6}{q^2 r^8})$.

9 d)

$9\ln p - 5\ln q + 7\ln r = \ln p^9 - \ln q^5 + \ln r^7$
$= \ln(\frac{p^9 r^7}{q^5})$.

10 a)

$2\ln p - (\ln q + 3\ln r) = \ln p^2 - (\ln q + \ln r^3)$
$= \ln p^2 - \ln(qr^3) = \ln(\frac{p^2}{qr^3})$.

10 b)

$\ln p - (2\ln q - \ln r) = \ln p - (\ln q^2 - \ln r)$
$= \ln p - \ln(\frac{q^2}{r}) = \ln(\frac{rp}{q^2})$.

10 c)

$3\ln p - 2(\ln q + 2\ln r) = \ln p^3 - 2(\ln q + \ln r^2)$
$= \ln p^3 - 2\ln(qr^2) = \ln p^3 - \ln(q^2 r^4) = \ln(\frac{p^3}{q^2 r^4})$.

10 d)

$4\ln p - 3(2\ln q - 3\ln r)$
$= \ln p^4 - 3(\ln q^2 - \ln r^3) = \ln p^4 - 3\ln(\frac{q^2}{r^3})$
$= \ln p^4 - \ln(\frac{q^6}{r^9}) = \ln(\frac{p^4 r^9}{q^6})$.

11 a)

$\ln x + \ln y = 5 \Rightarrow \ln(xy) = 5 \Rightarrow xy = e^5$.

11 b)

$\ln y - \ln x = 12 \Rightarrow \ln(\frac{y}{x}) = 12 \Rightarrow \frac{y}{x} = e^{12}$.

11 c)

$2\ln x - \ln y = 4 \Rightarrow \ln x^2 - \ln y = 4 \Rightarrow \ln(\frac{x^2}{y}) = 4$
$\Rightarrow \frac{x^2}{y} = e^4$.

11 d)

$4\ln x + 3\ln y = 8 \Rightarrow \ln x^4 + \ln y^3 = 8$
$\Rightarrow \ln(x^4y^3) = 8 \Rightarrow x^4y^3 = e^8$.

12)

$\ln x = 2\ln y \Rightarrow \ln x = \ln y^2 \Rightarrow x = y^2$
$7y = y^2 + 10 \Rightarrow y^2 - 7y + 10 = 0$
$\Rightarrow (y-5)(y-2) = 0 \Rightarrow y = 5$ or $y = 2$
When $y = 5$, $x = 5^2 = 25$
When $y = 2$, $x = 2^2 = 4$
The graphs intersect at $(4, 2)$ and $(25, 5)$.

13 a)

$e^{2x} - 16e^x = 0 \Rightarrow e^x(e^x - 16) = 0$
$e^x \neq 0$ so $e^x = 16 \Rightarrow x = \ln 16$ is the only real solution.

13 b)

$e^{3x} - 16e^x = 0 \Rightarrow e^x(e^{2x} - 16) = 0$
$e^x \neq 0$ so $e^{2x} = 16 \Rightarrow 2x = \ln 16 \Rightarrow x = \frac{1}{2}\ln 16$
$\Rightarrow x = \ln 4$ is the only real solution.

13 c)

$e^{3x} - 16e^{2x} = 0 \Rightarrow e^{2x}(e^x - 16) = 0$
$e^{2x} \neq 0$ so $e^x = 16 \Rightarrow x = \ln 16$ is the only real solution.

13 d)

$e^{4x} - 16e^{2x} = 0 \Rightarrow e^{2x}(e^{2x} - 16) = 0$
$e^{2x} \neq 0$ so $e^{2x} = 16 \Rightarrow 2x = \ln 16 \Rightarrow x = \frac{1}{2}\ln 16$
$\Rightarrow x = \ln 4$ is the only real solution.

14 a)

$e^x = e^{2x+1} \Rightarrow x = 2x + 1 \Rightarrow x = -1$.

14 b)

$e^{2x} = e^{3x-5} \Rightarrow 2x = 3x - 5 \Rightarrow x = 5$.

14 c)

$e^x = 2e^{4x} \Rightarrow \ln e^x = \ln(2e^{4x})$
$\Rightarrow \ln e^x = \ln 2 + \ln e^{4x} \Rightarrow x = \ln 2 + 4x$
$\Rightarrow 3x = -\ln 2 \Rightarrow x = -\frac{1}{3}\ln 2.$

14 d)

$e^{2x} = 3e^{x-1} \Rightarrow \ln e^{2x} = \ln(3e^{x-1})$
$\Rightarrow \ln e^{2x} = \ln 3 + \ln e^{x-1} \Rightarrow 2x = \ln 3 + x - 1$
$\Rightarrow x = \ln 3 - 1.$

15 a)

$4e^x + 27e^{-x} - 39 = 0 \Rightarrow 4(e^x)^2 + 27 - 39e^x = 0$
$\Rightarrow 4y^2 + 27 - 39y = 0 \Rightarrow 4y^2 - 39y + 27 = 0.$

15 b)

$4y^2 - 39y + 27 = 0 \Rightarrow (4y - 3)(y - 9) = 0$
$\Rightarrow y = \frac{3}{4}$ or $y = 9$
$e^x = \frac{3}{4} \Rightarrow x = \ln \frac{3}{4}$
$e^x = 9 \Rightarrow x = \ln 9.$

16 a)

$e^{2x} - 6e^x + 8 = 0 \Rightarrow (e^x)^2 - 6e^x + 8 = 0$
$y = e^x : y^2 - 6y + 8 = 0 \Rightarrow (y - 2)(y - 4) = 0$
$\Rightarrow y = 2$ or $y = 4$
$e^x = 2 \Rightarrow x = \ln 2$
$e^x = 4 \Rightarrow x = \ln 4.$

16 b)

$e^{2x} - 13e^x + 40 = 0 \Rightarrow (e^x)^2 - 13e^x + 40 = 0$
$y = e^x : y^2 - 13y + 40 = 0 \Rightarrow (y - 5)(y - 8) = 0$
$\Rightarrow y = 5$ or $y = 8$
$e^x = 5 \Rightarrow x = \ln 5$
$e^x = 8 \Rightarrow x = \ln 8.$

16 c)

$e^x + 4 - 45e^{-x} = 0 \Rightarrow e^{2x} + 4e^x - 45 = 0$
$\Rightarrow (e^x)^2 + 4e^x - 45 = 0$
$y = e^x : y^2 + 4y - 45 = 0 \Rightarrow (y + 9)(y - 5) = 0$
$\Rightarrow y = -9$ or $y = 5$
$e^x \neq -9$
$e^x = 5 \Rightarrow x = \ln 5.$

16 d)

$e^x + 5 - 150e^{-x} = 0 \Rightarrow e^{2x} + 5e^x - 150 = 0$
$\Rightarrow (e^x)^2 + 5e^x - 150 = 0$
$y = e^x : y^2 + 5y - 150 = 0 \Rightarrow (y + 15)(y - 10)$
$= 0 \Rightarrow y = -15$ or $y = 10$
$e^x \neq -15$
$e^x = 10 \Rightarrow x = \ln 10.$

17 a)

$2e^{2x} - 7e^x + 6 = 0 \Rightarrow 2(e^x)^2 - 7e^x + 6 = 0$
$y = e^x : 2y^2 - 7y + 6 = 0 \Rightarrow (2y - 3)(y - 2) = 0$
$\Rightarrow y = \frac{3}{2}$ or $y = 2$
$e^x = \frac{3}{2} \Rightarrow x = \ln \frac{3}{2}$
$e^x = 2 \Rightarrow x = \ln 2.$

17 b)

$4e^{2x} - 28e^x + 45 = 0 \Rightarrow 4(e^x)^2 - 28e^x + 45 = 0$
$y = e^x : 4y^2 - 28y + 45 = 0$
$\Rightarrow (2y - 5)(2y - 9) = 0 \Rightarrow y = \frac{5}{2}$ or $y = \frac{9}{2}$
$e^x = \frac{5}{2} \Rightarrow x = \ln \frac{5}{2}$
$e^x = \frac{9}{2} \Rightarrow x = \ln \frac{9}{2}.$

17 c)

$6e^x - 23 - 18e^{-x} = 0 \Rightarrow 6e^{2x} - 23e^x - 18 = 0$
$\Rightarrow 6(e^x)^2 - 23e^x - 18 = 0$
$y = e^x : 6y^2 - 23y - 18 = 0$
$\Rightarrow (3y + 2)(2y - 9) = 0 \Rightarrow y = -\frac{2}{3}$ or $y = \frac{9}{2}$
$e^x \neq -\frac{2}{3}$
$e^x = \frac{9}{2} \Rightarrow x = \ln \frac{9}{2}.$

17 d)

$8e^x + 14 - 15e^{-x} = 0 \Rightarrow 8e^{2x} + 14e^x - 15 = 0$
$\Rightarrow 8(e^x)^2 + 14e^x - 15 = 0$
$y = e^x : 8y^2 + 14y - 15 = 0$
$\Rightarrow (2y + 5)(4y - 3) = 0 \Rightarrow y = -\frac{5}{2}$ or $y = \frac{3}{4}$
$e^x \neq -\frac{5}{2}$
$e^x = \frac{3}{4} \Rightarrow x = \ln \frac{3}{4}.$

18)

$4^{x-3} = 5^{x+6} \Rightarrow \ln 4^{x-3} = \ln 5^{x+6}$
$\Rightarrow (x - 3) \ln 4 = (x + 6) \ln 5$
$\Rightarrow x \ln 4 - 3 \ln 4 = x \ln 5 + 6 \ln 5$
$\Rightarrow x \ln 4 - x \ln 5 = 6 \ln 5 + 3 \ln 4$
$\Rightarrow x(\ln 4 - \ln 5) = 6 \ln 5 + 3 \ln 4 \Rightarrow x = \frac{6 \ln 5 + 3 \ln 4}{\ln 4 - \ln 5}.$

19)

$\ln 5 \times \ln 125 \times \ln 625 = \ln 5 \times \ln 5^3 \times \ln 5^4$
$= \ln 5 \times 3 \ln 5 \times 4 \ln 5 = 12(\ln 5)^3 > 12$ as $\ln 5 > 1.$

20 a)

$2^x = 5^{x+4} \Rightarrow \ln 2^x = \ln 5^{x+4}$
$\Rightarrow x \ln 2 = (x + 4) \ln 5 \Rightarrow x \ln 2 = x \ln 5 + 4 \ln 5$
$\Rightarrow x \ln 2 - x \ln 5 = 4 \ln 5 \Rightarrow x(\ln 2 - \ln 5) = 4 \ln 5$
$\Rightarrow x = \frac{4 \ln 5}{\ln 2 - \ln 5}.$

20 b)

$4^{x-1} = 6^{x+2} \Rightarrow \ln 4^{x-1} = \ln 6^{x+2}$
$\Rightarrow (x-1)\ln 4 = (x+2)\ln 6$
$\Rightarrow x\ln 4 - \ln 4 = x\ln 6 + 2\ln 6$
$\Rightarrow x\ln 4 - x\ln 6 = 2\ln 6 + \ln 4$
$\Rightarrow x(\ln 4 - \ln 6) = 2\ln 6 + \ln 4 \Rightarrow x = \frac{2\ln 6 + \ln 4}{\ln 4 - \ln 6}$.

20 c)

$5^{3-x} = 7^{x+1} \Rightarrow \ln 5^{3-x} = \ln 7^{x+1}$
$\Rightarrow (3-x)\ln 5 = (x+1)\ln 7$
$\Rightarrow 3\ln 5 - x\ln 5 = x\ln 7 + \ln 7$
$\Rightarrow 3\ln 5 - \ln 7 = x\ln 5 + x\ln 7$
$\Rightarrow 3\ln 5 - \ln 7 = x(\ln 5 + \ln 7) \Rightarrow x = \frac{3\ln 5 - \ln 7}{\ln 5 + \ln 7}$.

20 d)

$8^{9-x} = 6^{8-x} \Rightarrow \ln 8^{9-x} = \ln 6^{8-x}$
$\Rightarrow (9-x)\ln 8 = (8-x)\ln 6$
$\Rightarrow 9\ln 8 - x\ln 8 = 8\ln 6 - x\ln 6$
$\Rightarrow x\ln 6 - x\ln 8 = 8\ln 6 - 9\ln 8$
$\Rightarrow x(\ln 6 - \ln 8) = 8\ln 6 - 9\ln 8 \Rightarrow x = \frac{8\ln 6 - 9\ln 8}{\ln 6 - \ln 8}$.

21 a)

$2\ln(3x-5) - 4 = 0 \Rightarrow \ln(3x-5) = 2$
$\Rightarrow 3x - 5 = e^2 \Rightarrow x = \frac{e^2+5}{3}$.

21 b)

$2\ln(3-2x) - 9 = 0 \Rightarrow \ln(3-2x) = \frac{9}{2}$
$\Rightarrow 3 - 2x = e^{\frac{9}{2}} \Rightarrow x = \frac{3 - e^{\frac{9}{2}}}{2}$.

21 c)

$2\ln(x+4) - \ln 5 = 0 \Rightarrow \ln(x+4)^2 = \ln 5$
$\Rightarrow (x+4)^2 = 5 \Rightarrow x + 4 = \pm\sqrt{5} \Rightarrow x = -4 \pm \sqrt{5}$
$x \neq -4 - \sqrt{5}$ so $x = -4 + \sqrt{5}$.

21 d)

$5\ln 2x - 3\ln 3x = 1 \Rightarrow \ln(32x^5) - \ln(27x^3) = 1$
$\Rightarrow \ln(\frac{32x^5}{27x^3}) = 1 \Rightarrow \ln(\frac{32}{27}x^2) = 1 \Rightarrow \frac{32}{27}x^2 = e$
$\Rightarrow x^2 = \frac{27}{32}e \Rightarrow x = \pm\sqrt{\frac{27}{32}e}$
$x \neq -\sqrt{\frac{27}{32}}$ so $x = \sqrt{\frac{27}{32}e}$.

22)

$2\ln x - \ln(x+12) = 0 \Rightarrow \ln x^2 - \ln(x+12) = 0$
$\Rightarrow \ln(\frac{x^2}{x+12}) = 0 \Rightarrow \frac{x^2}{x+12} = 1 \Rightarrow x^2 = x + 12$
$\Rightarrow x^2 - x - 12 = 0 \Rightarrow (x-4)(x+3) = 0$
$\Rightarrow x = 4$ or $x = -3$
$x \neq -3$ so $x = 4$.

23 a)

$\ln x = 2 + \ln(x+3) \Rightarrow \ln x = \ln e^2 + \ln(x+3)$
$\Rightarrow \ln x = \ln(e^2(x+3)) = x = e^2(x+3)$
$\Rightarrow x = e^2x + 3e^2 \Rightarrow x - e^2x = 3e^2$
$\Rightarrow x(1-e^2) = 3e^2 = x = \frac{3e^2}{1-e^2}$.

23 b)

$\ln(x-1) = 3 + \ln(x+1)$
$\Rightarrow \ln(x-1) = \ln e^3 + \ln(x+1)$
$\Rightarrow \ln(x-1) = \ln(e^3(x+1)) \Rightarrow x - 1 = e^3(x+1)$
$\Rightarrow x - 1 = e^3x + e^3 \Rightarrow x - e^3x = e^3 + 1$
$\Rightarrow x(1-e^3) = e^3 + 1 \Rightarrow x = \frac{e^3+1}{1-e^3}$.

23 c)

$\ln 2x = 4 - \ln x \Rightarrow \ln 2x = \ln e^4 - \ln x$
$\Rightarrow \ln 2x = \ln \frac{e^4}{x} \Rightarrow 2x = \frac{e^4}{x} \Rightarrow 2x^2 = e^4 \Rightarrow x^2 = \frac{e^4}{2}$
$\Rightarrow x = \sqrt{\frac{e^4}{2}} = \frac{e^2}{\sqrt{2}}$.

23 d)

$\ln 4x = 5 - \ln(x+1) \Rightarrow \ln 4x = \ln e^5 - \ln(x+1)$
$\Rightarrow \ln 4x = \ln \frac{e^5}{x+1} \Rightarrow 4x = \frac{e^5}{x+1} \Rightarrow 4x(x+1) = e^5$
$\Rightarrow 4x^2 + 4x - e^5 = 0 \Rightarrow x = \frac{-4+\sqrt{16+16e^5}}{8} = \frac{-1+\sqrt{1+e^5}}{2}$.

24)

$\ln(x^2+2) = 4\ln x \Rightarrow \ln(x^2+2) = \ln x^4$
$\Rightarrow x^2 + 2 = x^4 \Rightarrow x^4 - x^2 - 2 = 0$
$\Rightarrow (x^2)^2 - x^2 - 2 = 0$
$u = x^2 : u^2 - u - 2 = 0 \Rightarrow (u-2)(u+1) = 0$
$\Rightarrow u = 2$ or $u = -1$
$x^2 = 2 \Rightarrow x = \pm\sqrt{2}$
$x^2 \neq -1$
$x \neq -\sqrt{2}$ so $x = \sqrt{2}$
When $x = \sqrt{2}, y = 4\ln(\sqrt{2}) = 4\ln 2^{\frac{1}{2}} = 2\ln 2 = \ln 4$
So the curves intersect at $(\sqrt{2}, 4\ln\sqrt{2}) = (\sqrt{2}, \ln 4)$.

25 a)

$2\ln x = 2 + \ln 2x \Rightarrow \ln x^2 = \ln e^2 + \ln 2x$
$\Rightarrow \ln x^2 = \ln(2e^2x) \Rightarrow x^2 = 2e^2x$
$\Rightarrow x = 2e^2$ as $x \neq 0$.

25 b)

$2\ln x = 3 - \ln 8x \Rightarrow \ln x^2 = \ln e^3 - \ln 8x$
$\Rightarrow \ln x^2 = \ln \frac{e^3}{8x} \Rightarrow x^2 = \frac{e^3}{8x} \Rightarrow x^3 = \frac{e^3}{8} \Rightarrow x = \frac{e}{2}$.

25 c)

$2\ln x = \ln 4x - 1 \Rightarrow \ln x^2 = \ln 4x - \ln e$
$\Rightarrow \ln x^2 = \ln(\frac{4x}{e}) \Rightarrow x^2 = \frac{4x}{e} \Rightarrow x = \frac{4}{e}$.

25 d)

$2\ln x = \ln 10x - 4 \Rightarrow \ln x^2 = \ln 10x - \ln e^4$
$\Rightarrow \ln x^2 = \ln(\frac{10x}{e^4}) \Rightarrow x^2 = \frac{10x}{e^4} \Rightarrow x = \frac{10}{e^4}$.

26)

$3\ln x = 3 - \ln 16x \Rightarrow \ln x^3 = \ln e^3 - \ln 16x$
$\Rightarrow \ln x^3 = \ln(\frac{e^3}{16x}) \Rightarrow x^3 = \frac{e^3}{16x} \Rightarrow 16x^4 = e^3$
$\Rightarrow x^4 = \frac{1}{16}e^3 \Rightarrow x = \pm\frac{1}{2}e^{\frac{3}{4}}$
$x \neq -\frac{1}{2}e^{\frac{3}{4}}$ so $x = \frac{1}{2}e^{\frac{3}{4}}$ only.

27 a)

$\ln x + \frac{32}{\ln x} = 12 \Rightarrow (\ln x)^2 + 32 = 12\ln x$
$\Rightarrow (\ln x)^2 - 12\ln x + 32 = 0$
$\Rightarrow (\ln x - 8)(\ln x - 4) = 0$
$\ln x = 8 \Rightarrow x = e^8$
$\ln x = 4 \Rightarrow x = e^4$.

27 b)

$2\ln x + \frac{6}{\ln x} = 7 \Rightarrow 2(\ln x)^2 + 6 = 7\ln x$
$\Rightarrow 2(\ln x)^2 - 7\ln x + 6 = 0$
$\Rightarrow (2\ln x - 3)(\ln x - 2) = 0$
$\ln x = \frac{3}{2} \Rightarrow x = e^{\frac{3}{2}}$
$\ln x = 2 \Rightarrow x = e^2$.

27 c)

$6\ln x + 11 = \frac{10}{\ln x} \Rightarrow 6(\ln x)^2 + 11\ln x = 10$
$\Rightarrow 6(\ln x)^2 + 11\ln x - 10 = 0$
$\Rightarrow (2\ln x + 5)(3\ln x - 2) = 0$
$\ln x = -\frac{5}{2} \Rightarrow x = e^{-\frac{5}{2}}$
$\ln x = \frac{2}{3} \Rightarrow x = e^{\frac{2}{3}}$.

27 d)

$10\ln x - 33 = \frac{7}{\ln x} \Rightarrow 10(\ln x)^2 - 33\ln x = 7$
$\Rightarrow 10(\ln x)^2 - 33\ln x - 7 = 0$
$\Rightarrow (5\ln x + 1)(2\ln x - 7) = 0$
$\ln x = -\frac{1}{5} \Rightarrow x = e^{-\frac{1}{5}}$
$\ln x = \frac{7}{2} \Rightarrow x = e^{\frac{7}{2}}$.

28)

$2\ln(x+10) = \ln x + \ln(2x-1)$
$\Rightarrow \ln((x+10)^2) = \ln(x(2x-1))$
$\Rightarrow (x+10)^2 = x(2x-1)$
$\Rightarrow x^2 + 20x + 100 = 2x^2 - x$
$\Rightarrow x^2 - 21x - 100 = 0$
$\Rightarrow (x-25)(x+4) = 0$
$\Rightarrow x = 25$ as $x \neq -4$.

29)

$\ln(5-3x)+\ln(8+4x)=2\ln(x+2)$
$\Rightarrow \ln((5-3x)(8+4x))=\ln((x+2)^2)$
$\Rightarrow (5-3x)(8+4x)=(x+2)^2$
$\Rightarrow -12x^2-4x+40=x^2+4x+4$
$\Rightarrow 13x^2+8x-36=0 \Rightarrow (13x-18)(x+2)=0$
$x\neq -2$ so $x=\frac{18}{13}$.

30)

$2\ln(5-2x)=\ln 2+\ln(\frac{233}{8}-2(x-\frac{5}{4})^2)$
$\Rightarrow \ln(5-2x)^2=\ln(\frac{233}{4}-4(x-\frac{5}{4})^2)$
$\Rightarrow \ln(5-2x)^2=\ln(\frac{233}{4}-4x^2+10x-\frac{25}{4})$
$\Rightarrow \ln(5-2x)^2=\ln(52-4x^2+10x)$
$\Rightarrow (5-2x)^2=52-4x^2+10x$
$\Rightarrow 25-20x+4x^2=52-4x^2+10x$
$\Rightarrow 8x^2-30x-27=0 \Rightarrow (4x+3)(2x-9)=0$
$\Rightarrow x=-\frac{3}{4}$ or $x=\frac{9}{2}$
However, $\ln(5-2\times\frac{9}{2})=\ln(-4)$, so as logarithms of negative numbers are not real, the only real solution to the original question is $x=-\frac{3}{4}$.

31 a)

$1-28e^{-3x}-4e^{-2x}+7e^{-x}=0$
$\Rightarrow e^{3x}-28-4e^x+7e^{2x}=0$
$\Rightarrow (e^x)^3+7(e^x)^2-4e^x-28=0$
$\Rightarrow y^3+7y^2-4y-28=0.$

31 b)

$y^3+7y^2-4y-28=0$
$\Rightarrow (y+7)(y+2)(y-2)=0 \Rightarrow y=-7$ or $y=-2$ or $y=2$
$e^x\neq -7$
$e^x\neq -2$
$e^x=2\Rightarrow x=\ln 2.$

32)

$y^2-3y+2=0\Rightarrow (y-2)(y-1)=0$
$\Rightarrow y=2$ or $y=1$
When $y=2$, $2=\ln x\Rightarrow x=e^2$
When $y=1$, $1=\ln x\Rightarrow x=e$.

33)

$6(\ln x)^2-19\ln x+15=0$
$y=\ln x: 6y^2-19y+15=0$
$\Rightarrow (2y-3)(3y-5)=0\Rightarrow y=\frac{3}{2}$ or $y=\frac{5}{3}$
When $y=\frac{3}{2}$, $\frac{3}{2}=\ln x\Rightarrow x=e^{\frac{3}{2}}$
When $y=\frac{5}{3}$, $\frac{5}{3}=\ln x\Rightarrow x=e^{\frac{5}{3}}$.

34)

$\ln 2x + \ln 2x^2 + \ln 2x^3 = 6$
$\Rightarrow \ln(2x \times 2x^2 \times 2x^3) = 6$
$\Rightarrow \ln(8x^6) = 6 \Rightarrow 8x^6 = e^6$
$\Rightarrow x^6 = \frac{1}{8}e^6 \Rightarrow x = \frac{1}{\sqrt{2}}e$
$x \neq -\frac{1}{\sqrt{2}}e$ as $\ln(2 \times -\frac{1}{\sqrt{2}}e)$ is not a real solution.

35)

$\frac{e^x}{3^x} = e \Rightarrow \ln(\frac{e^x}{3^x}) = 1 \Rightarrow \ln e^x - \ln 3^x = 1$
$\Rightarrow x - x\ln 3 = 1 \Rightarrow x(1 - \ln 3) = 1 \Rightarrow x = \frac{1}{1-\ln 3}$.

36)

$3^x e^{2x+3} = 2 \Rightarrow \ln(3^x e^{2x+3}) = \ln 2$
$\Rightarrow \ln 3^x + \ln e^{2x+3} = \ln 2 \Rightarrow x\ln 3 + 2x + 3 = \ln 2$
$\Rightarrow x(\ln 3 + 2) = \ln 2 - 3 \Rightarrow x = \frac{\ln 2 - 3}{\ln 3 + 2} = \frac{-3+\ln 2}{2+\ln 3}$.

37)

$5^x e^{8x} = e^{10} \Rightarrow \ln(5^x e^{8x}) = \ln e^{10}$
$\Rightarrow \ln 5^x + \ln e^{8x} = 10 \Rightarrow x\ln 5 + 8x = 10$
$\Rightarrow x(\ln 5 + 8) = 10 \Rightarrow x = \frac{10}{\ln 5 + 8} = \frac{10}{8+\ln 5}$.

38)

$\frac{3x^2-19x+6}{x^2-x-30} = \frac{(3x-1)(x-6)}{(x+5)(x-6)} = \frac{3x-1}{x+5}$
$\ln(3x^2 - 19x + 6) - 1 = \ln(x^2 - x - 30)$
$\Rightarrow \ln(3x^2 - 19x + 6) - \ln(x^2 - x - 30) = 1$
$\Rightarrow \ln(\frac{3x^2-19x+6}{x^2-x-30}) = 1 \Rightarrow \ln(\frac{3x-1}{x+5}) = 1 \Rightarrow \frac{3x-1}{x+5} = e$
$\Rightarrow 3x - 1 = e(x + 5) \Rightarrow 3x - 1 = ex + 5e$
$\Rightarrow 3x - ex = 1 + 5e \Rightarrow x(3 - e) = 1 + 5e$
$\Rightarrow x = \frac{1+5e}{3-e}$.

39)

$2e^x - 11 = 40e^{-x} \Rightarrow 2(e^x)^2 - 11e^x = 40$
$\Rightarrow 2(e^x)^2 - 11e^x - 40 = 0$
Let $y = e^x$: $2y^2 - 11y - 40 = 0$
$\Rightarrow (2y + 5)(y - 8) = 0 \Rightarrow y = -\frac{5}{2}$ or $y = 8$
$e^x \neq -\frac{5}{2}$
$e^x = 8 \Rightarrow x = \ln 8$
$y|_{x=\ln 8} = 40e^{-\ln 8} = 40e^{\ln \frac{1}{8}} = 40 \times \frac{1}{8} = 5$
$\therefore$ the curves intersect at $(\ln 8, 5)$.

40)

$(\ln 2, 12) : 12 = ae^{\ln 2} + b \Rightarrow 12 = 2a + b \quad (1)$
$(-3\ln 2, \frac{87}{16}) : \frac{87}{16} = ae^{-3\ln 2} + b \Rightarrow \frac{87}{16} = ae^{\ln 2^{-3}} + b$
$\Rightarrow \frac{87}{16} = \frac{1}{8}a + b \quad (2)$
$(1) - (2) \Rightarrow \frac{105}{16} = \frac{15}{8}a \Rightarrow a = \frac{7}{2}$
Sub into (1): $12 = 2 \times \frac{7}{2} + b \Rightarrow b = 5$.

41)

$(e^{12}, -\frac{55}{2}) : -\frac{55}{2} = p\ln(e^{12}) + q$
$\Rightarrow -\frac{55}{2} = 12p + q \quad (1)$
$(e, \frac{11}{6}) : \frac{11}{6} = p\ln e + q \Rightarrow \frac{11}{6} = p + q \quad (2)$
$(1) - (2) \Rightarrow -\frac{88}{3} = 11p \Rightarrow p = -\frac{8}{3}$
Sub into (1): $-\frac{55}{2} = 12 \times (-\frac{8}{3}) + q \Rightarrow q = \frac{9}{2}$.

42)

$2\ln y = x \Rightarrow \ln y^2 = x$
$e^{\ln y^2} + y^2 = 4 \Rightarrow y^2 + y^2 = 4 \Rightarrow 2y^2 = 4$
$\Rightarrow y^2 = 2 \Rightarrow y = \pm\sqrt{2}$
$y \neq -\sqrt{2}$ so $y = \sqrt{2}$
$\therefore x = \ln(\sqrt{2})^2 = \ln 2.$

43)

$x + e^y = 4 \Rightarrow e^y = 4 - x \Rightarrow y = \ln(4 - x)$
$\ln((x-4)^2) - \ln(4 - x) = 2 \Rightarrow \ln(\frac{(x-4)^2}{4-x}) = 2$
$\Rightarrow \ln(\frac{(4-x)^2}{4-x}) = 2 \Rightarrow \ln(4 - x) = 2 \Rightarrow 4 - x = e^2$
$\Rightarrow x = 4 - e^2$
When $x = 4 - e^2$, $y = \ln(4 - (4 - e^2)) = \ln e^2 = 2$
So the curves intersect at $(4 - e^2, 2)$.

44 a)

$t = 0 \Rightarrow h = 20 - 5\ln(0 + 1) = 20$ metres.

44 b)

$t = 1 \Rightarrow h = 20 - 5\ln(1 + 1) = 16.5$ metres to 3sf.

44 c)

$10 = 20 - 5\ln(t + 1) \Rightarrow 5\ln(t + 1) = 10$
$\Rightarrow \ln(t + 1) = 2 \Rightarrow t + 1 = e^2$
$\Rightarrow t = e^2 - 1 = 6.39$ hours to 3sf.

45 a)

$t = 0 \Rightarrow P = 100\ln(\frac{0+3}{3}) + 500 = 500$ hawks.

45 b)

$t = 5 \Rightarrow P = 100\ln(\frac{5+3}{3}) + 500 \approx 598$ hawks.

45 c)

$700 = 100\ln(\frac{t+3}{3}) + 500 \Rightarrow 100\ln(\frac{t+3}{3}) = 200$
$\Rightarrow \ln(\frac{t+3}{3}) = 2 \Rightarrow \frac{t+3}{3} = e^2$
$\Rightarrow t = 3e^2 - 3 = 19.16\ldots$ years
So after 20 years, the population will have reached 700.

46 a)

$t = 0 \Rightarrow T = 2\ln(0+3) - \ln(0+2) + \ln 100$
$= 6.11°\text{C}$ to 3sf.

46 b)

$\ln 3000 = 2\ln(t+3) - \ln(t+2) + \ln 100$
$\Rightarrow \ln 3000 = \ln((t+3)^2) - \ln(t+2) + \ln 100$
$\Rightarrow \ln 3000 = \ln(\frac{(t+3)^2}{t+2}) + \ln 100$
$\Rightarrow \ln 3000 = \ln(\frac{100(t+3)^2}{t+2})$
$\Rightarrow 3000 = \frac{100(t+3)^2}{t+2}$
$\Rightarrow 3000(t+2) = 100(t+3)^2$
$\Rightarrow 30(t+2) = (t+3)^2$
$\Rightarrow 30t + 60 = t^2 + 6t + 9$
$\Rightarrow t^2 - 24t - 51 = 0 \Rightarrow t = 12 \pm \sqrt{195}$
$t \neq 12 - \sqrt{195}$ so $t = 12 + \sqrt{195} = 25.96\ldots$
So after 26 days, the temperature will have reached $(\ln 3000)°\text{C}$.